High school · reasoning

Parity and invariants

Progressive exercises

Exercise 1

Start at 7 and add or subtract 4 each move. Can you reach 20? What about 23?

Exercise 2

Five numbers, each 1 or -1, have product -1. Prove that changing two signs per move cannot make all five positive.

Exercise 3

Replace two integers a and b by $a+b$. What invariant does this preserve?

Exercise 4

Why need two states of the same parity not be connected by allowed moves? Give an example.